SAMPLE COURSE HANDBOOK - CST 3XX: APPLIED MACHINE LEARNING (B.Tech CSE, Semester 6)

This handbook is an illustrative teaching sample prepared for the Retrieval-Augmented Generation workshop. It describes the course structure, syllabus modules, examination pattern, internal evaluation scheme, laboratory guidelines and academic integrity rules for a third-year Computer Science and Engineering elective. Students should use it to test how a RAG system retrieves precise facts from a long document.

1. Course Objectives

The course introduces supervised and unsupervised learning, model evaluation, and the engineering of modern language-model applications. On completion, students will be able to formulate a learning problem, select an appropriate algorithm, measure generalisation error, and deploy a small retrieval-augmented question answering service. The course carries 4 credits with 3 lecture hours and 1 tutorial hour per week.

2. Syllabus - Module 1: Foundations of Learning

Types of learning: supervised, unsupervised, reinforcement. Hypothesis space, inductive bias, bias-variance trade-off. Linear regression with ordinary least squares and gradient descent. Regularisation using L1 (lasso) and L2 (ridge) penalties. Train, validation and test splits; k-fold cross validation. (9 hours)

3. Syllabus - Module 2: Classification

Logistic regression and the sigmoid function. Decision trees, entropy and information gain. Support vector machines, margin maximisation and the kernel trick. k-nearest neighbours and distance metrics including Euclidean, Manhattan and cosine distance. Naive Bayes classifier. (9 hours)

4. Syllabus - Module 3: Neural Networks and Representations

Perceptron, multilayer perceptron, backpropagation. Activation functions: ReLU, tanh, softmax. Word embeddings: one-hot encoding versus dense vectors, word2vec intuition,

sentence embeddings. Vector spaces, dot products, cosine similarity and why normalised vectors make cosine similarity equal to the dot product. (9 hours)

5. Syllabus - Module 4: Unsupervised Learning and Vector Search

k-means clustering, elbow method, hierarchical clustering. Principal component analysis for dimensionality reduction. Approximate nearest neighbour search: flat indexes, inverted file indexes (IVF), product quantisation and HNSW graphs. Introduction to the FAISS library and vector databases. (9 hours)

6. Syllabus - Module 5: Large Language Model Applications

Transformer architecture overview, tokenisation and context windows. Prompt engineering. Retrieval-Augmented Generation: document loading, chunking with overlap, embedding, indexing, retrieval, grading and grounded generation. Agentic workflows with state machines, query rewriting and self-correction. Hallucination and evaluation of RAG systems. (9 hours)

7. Examination Pattern - End Semester Examination (ESE)

The End Semester Examination is conducted for 100 marks with a duration of 3 hours. Part A contains 10 short-answer questions, two from each module, each carrying 3 marks, for a total of 30 marks. All questions in Part A are compulsory. Part B contains two questions from each module, of which the student must answer one full question per module. Each Part B question carries 14 marks and may contain a maximum of two sub-divisions, for a total of 70 marks. Programmable calculators are not permitted.

8. Continuous Internal Evaluation (CIE) - Internal Marks

Continuous Internal Evaluation carries 50 marks and is distributed as follows: Attendance 10 marks; Continuous Assessment Tests 25 marks; Assignments, quizzes and course project 15 marks. Two continuous assessment tests are conducted, the first after completing two and a half modules and the second after the remaining modules. The average of the two test scores is taken. A student must secure at least 40 percent of the CIE marks to be eligible to appear for the End Semester Examination.

9. Grading and Pass Criteria

The total mark for the course is 150 (50 CIE + 100 ESE). A student passes the course by securing at least 40 percent in the End Semester Examination and at least 45 percent of the aggregate of CIE and ESE marks. Letter grades are awarded on a 10-point scale: S (90 percent and above), A+ (85-89), A (80-84), B+ (70-79), B (60-69), C (50-59), P (45-49) and F (below 45).

10. Laboratory and Course Project Guidelines

Each student builds a mini project that loads a PDF corpus, splits it into chunks of roughly 500 to 1000 characters with 10 to 20 percent overlap, embeds the chunks, stores them in a FAISS index, and answers questions with citations to page numbers. The project report must include a table of retrieval distances for at least five test queries and a screenshot of the agent state graph. Projects are evaluated through a live demonstration and a viva voce.

11. Academic Integrity

Plagiarism in assignments, test answers or project code results in zero marks for the component concerned. Use of AI assistants must be declared in the project report, including the prompts used. Sharing API keys in public repositories is treated as a security violation and the keys must be revoked immediately.

12. Reference Books

1. Tom M. Mitchell, Machine Learning, McGraw Hill. 2. Christopher Bishop, Pattern Recognition and Machine Learning, Springer. 3. Aurelien Geron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O’Reilly. 4. Jay Alammar and Maarten Grootendorst, Hands-On Large Language Models, O’Reilly.